

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO1: *Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems.(BL2,BL3)*

Assessment Tool Weightage: 30%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Case Study

Duration: 2hrs

1. A logistics company is developing an RL-based warehouse robot that learns to pick, pack, and transport items efficiently while avoiding obstacles and minimizing delays. Analyze how the components of a Markov Decision Process (states, actions, rewards, policy, and environment) can be defined for this system. Design a reward function that balances speed, accuracy, and safety, and evaluate how reward design influences the robot's learning and behavior.

ANSWER:

A logistics company aims to develop a Reinforcement Learning (RL)-based warehouse robot that can pick, pack, and transport items efficiently while avoiding obstacles and minimizing delays. This problem can be modeled as a Markov Decision Process (MDP) because the robot repeatedly observes its environment, takes actions, receives rewards, and learns the best policy through interaction.

1. MDP Components

a) States (S)

A state represents the current condition of the robot and the warehouse.

Examples include:

- **Robot's current position**
- **Item location**
- **Destination location**
- **Battery level**
- **Presence of obstacles**
- **Whether the item has been picked or delivered**

Example State:

Robot at Shelf A, Item Not Picked, Battery = 80%, Obstacle Detected.

b) Actions (A)

Actions are the operations the robot can perform.

Examples:

- **Move Forward**
- **Move Backward**
- **Turn Left**
- **Turn Right**
- **Pick Item**
- **Drop Item**
- **Stop**
- **Recharge Battery**

The selected action changes the robot's current state.

c) Environment (E)

The environment is the warehouse where the robot operates.

It includes:

- **Storage shelves**
- **Packing stations**

- **Delivery locations**
- **Obstacles**
- **Other robots**
- **Charging stations**

The environment responds to the robot's actions by providing a new state and a reward.

d) Reward Function (R)

The reward function guides the robot toward desirable behavior.

Action	Reward
Correctly pick an item	+20
Successfully deliver an item	+50
Reach destination quickly	+10
Safe navigation	+5
Collision with obstacle	-30
Wrong item picked	-20
Delivery delay	-10
Battery exhausted	-50

This reward function balances speed, accuracy, and safety.

e) Policy (π)

A policy defines the action the robot should take in each state.

Example:

- **If an obstacle is detected → Turn Left or Right.**
- **If the robot reaches the required shelf → Pick the item.**
- **If the robot reaches the destination → Drop the item.**
- **If the battery is low → Move to the charging station.**

The objective of the policy is to maximize cumulative rewards.

Reward Function Design

A balanced reward function can be expressed as:

where:

- **Speed reward encourages faster task completion.**
 - **Accuracy reward encourages picking and delivering the correct item.**
 - **Safety reward encourages obstacle avoidance.**
 - **Collision penalty discourages unsafe movements.**
 - **Delay penalty discourages inefficient paths.**
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Influence of Reward Design

The design of the reward function significantly affects the robot's learning and behavior.

- **If speed is rewarded too highly, the robot may rush and collide with obstacles.**
- **If safety receives excessive reward, the robot may move very slowly, reducing efficiency.**
- **If accuracy is ignored, the robot may deliver incorrect items.**
- **A balanced reward function helps the robot learn the optimal policy by encouraging fast, safe, and accurate task completion.**

Thus, proper reward design ensures that the robot achieves high productivity while maintaining safety and operational accuracy.

Conclusion

By modeling the warehouse robot as a Markov Decision Process, the logistics company can effectively apply Reinforcement Learning. Clearly defining the states, actions, environment, rewards, and policy enables the robot to learn optimal behavior through experience. A well-balanced reward function improves learning efficiency, minimizes delays and collisions, and ensures accurate item handling, leading to enhanced warehouse performance.

Code of Conduct:

*I Kandukuri.Kavya-192425283 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Kandukuri.Kavya-192425283

RUBRICS FOR CASESTUDY

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand